CS1 Week 7

Proportional Feedback, Root Locus

Recap Last Week

Quiz 1

Which factor appears in the Bode-form of a transfer function for a real zero at z?

A. $(s - z)$

B. $\left(\frac{s}{-z} + 1\right)$

C. $\frac{1}{s - z}$

D. $\frac{\text{Bode}}{\text{los}}$

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C. $\frac{1}{s - z}$

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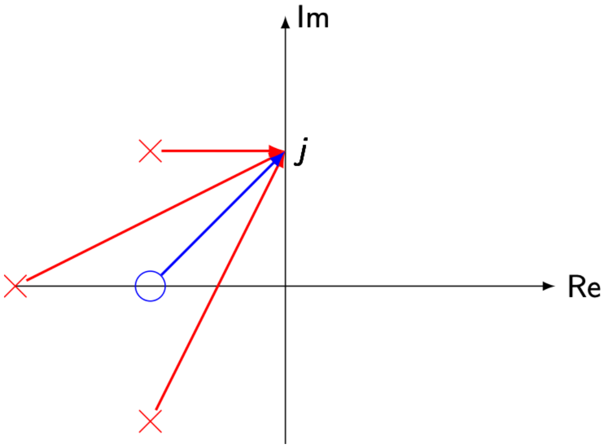
$$G(s) = \frac{k_{bode}}{s^q} \frac{\left(\frac{s}{-z_1} + 1\right)\left(\frac{s}{-z_2} + 1\right) \dots \left(\frac{s}{-z_m} + 1\right)}{\left(\frac{s}{-p_1} + 1\right)\left(\frac{s}{-p_2} + 1\right) \dots \left(\frac{s}{-p_{n-q}} + 1\right)}$$

Quiz 2

How do you compute $|G(j\omega)|$ from a pole–zero plot?



A. Sum the distances from $j\omega$ to all zeros and subtract the distances to all poles

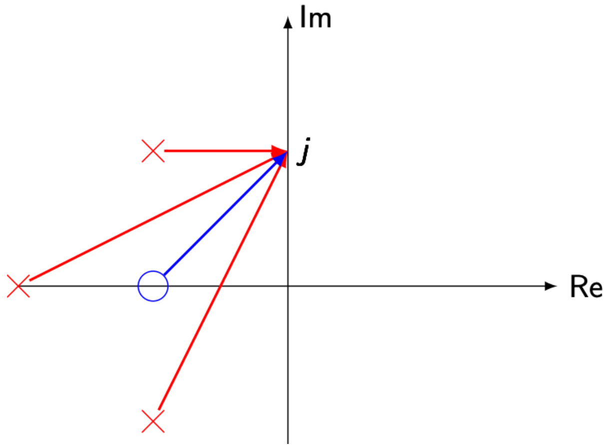
B. Product of distances from $j\omega$ to zeros divided by product of distances to poles

C. Count poles, subtract zeros, take absolute value

D. Measure with a physical ruler on your screen

Quiz 2

How do you compute $|G(j\omega)|$ from a pole–zero plot?



$|a \cdot b| = |a| \cdot |b|$

A. Sum the distances from $j\omega$ to all zeros and subtract the distances to all poles

B. Product of distances from $j\omega$ to zeros divided by product of distances to poles

C. Count poles, subtract zeros, take absolute value

D. Measure with a physical ruler on your screen

Quiz 3

What is the physical meaning of the poles of an LTI system?

A. They only set the steady-state (DC) gain

B. They quantify how much power the controller injects into the plant

C. They determine stability of the system

D. They determine the decay/growth rates and oscillation frequencies of the system's natural response

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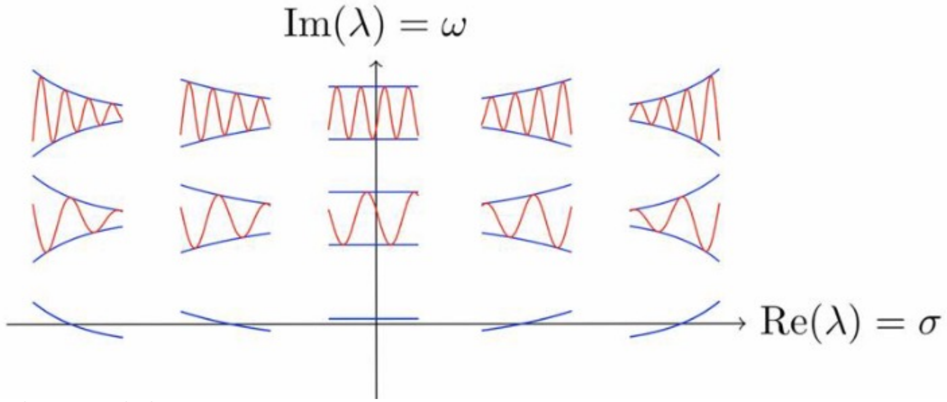
Poles = Eigenvalues of A!

$$\longrightarrow x(t) = c_1 e^{\lambda_1 t} v_1 + c_2 e^{\lambda_2 t} v_2$$



Quiz 4

What are non-minimum phase zeros?

A. Zeros in the right-half plane

B. The roots of the denominator of $G(s)$

C. They create an anticipatory effect and reduce phase lag

D. They cause an inverse initial response: the output first moves in the wrong direction before heading to the final value

Quiz 4

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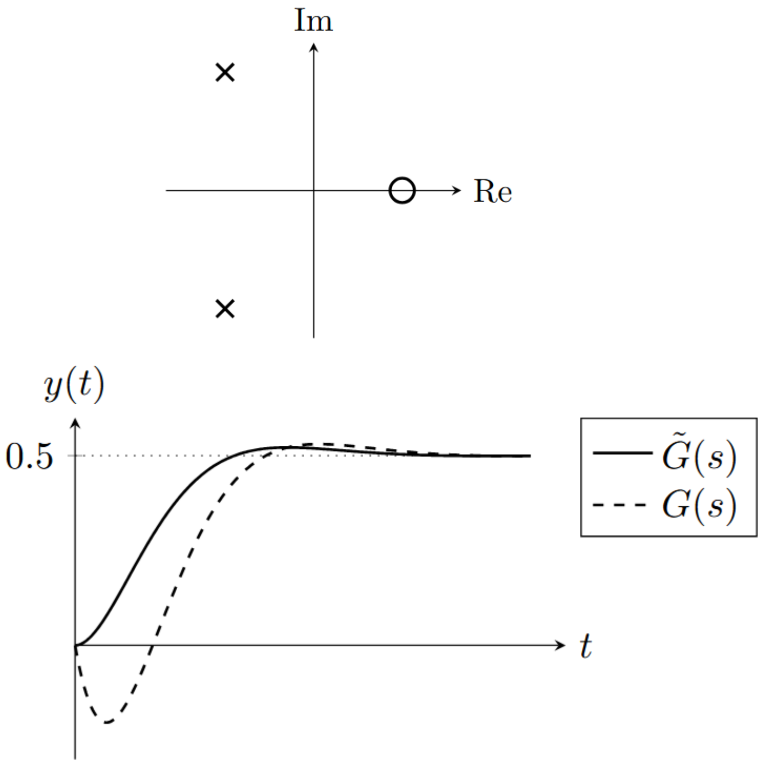

Quiz 5

Which statements about pole-zero cancellation are true?

A. Pole-Zero cancellation is always safe

B. Cancelling an unstable pole can make I/O look stable while internals stay unstable.

C. A zero near a pole partially cancels it, weakening that mode in the I/O response

D. Pole–zero proximity always improves robustness

Quiz 5

Which statements about pole-zero cancellation are true?

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B. Cancelling an unstable pole can make I/O look stable while internals stay unstable.

C. A zero near a pole partially cancels it, weakening that mode in the I/O response

D. Pole–zero proximity always improves robustness

Motivation

system modeling

Plant P

Classification & Linearization

LTI

state-space representation

$$\dot{x}(t) = Ax(t) + Bu(t)$$
$$y(t) = Cx(t) + Du(t)$$

Analysis

Time Response: System evolution in time
Stability: Output bounded or diverging
Transfer Function: I/O relation in s-domain
Poles & Zeros: Shape system dynamics

Synthesis

Controller C

What we looked at so far

Control Systems

What we are going to look at today

Course Schedule

	Subject	Week
Modeling	Introduction, Control Architectures, Motivation	1
	Modeling, Model examples	2
	System properties, Linearization	3
Analysis	Analysis: Time response, Stability	4
	Transfer functions 1: Definition and properties	5
	Transfer functions 2: Poles and Zeros	6
	Proportional feedback control, Root Locus	7
	Time-Domain specifications, PID control, Computer implementation	8
	Frequency response, Bode plots	9
	The Nyquist condition, Time delays	10
Synthesis	Frequency-domain Specifications, Dynamic Compensation, Loop Shaping	11
	Time delays, Successive loop closure, Nonlinearities	12
	Describing functions	13
	Intro to Uncertainty and Robustness	14

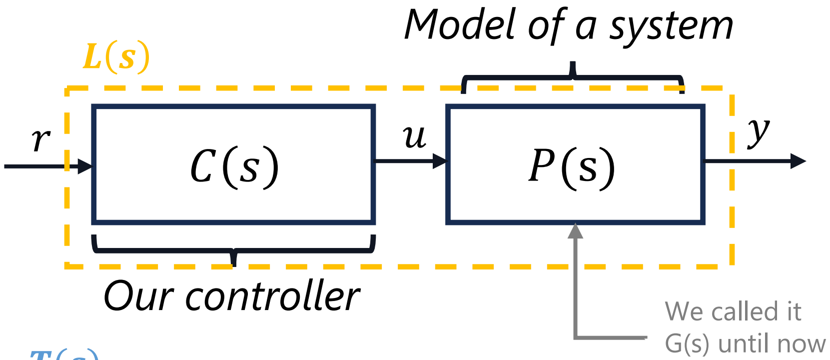
Today

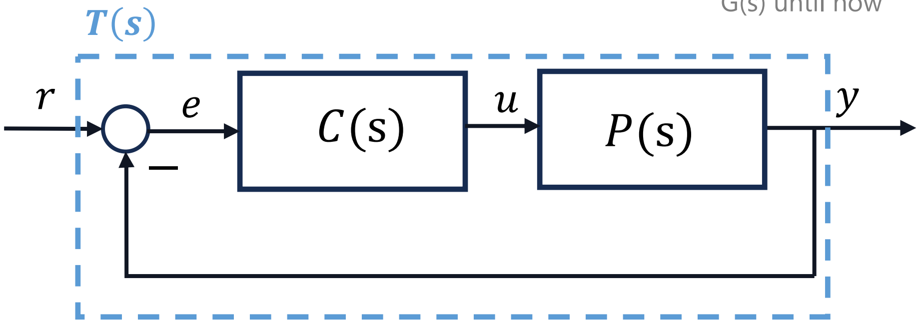
1. Feedback Control
2. Root-Locus

1. Feedback Control

The Initial Problem

- Until now, we only knew the transfer function $P(s) = G(s)$, describing how the system reacts to an input $u(t)$
- But once we add feedback, the controller influences the loop, we must now look at the entire interaction:
- The open-loop Transfer Function r $\longrightarrow y$ is given by $\boldsymbol{L(s) = C(s)P(s)}$





The closed-loop TF r $\longrightarrow y$ is derived like this:

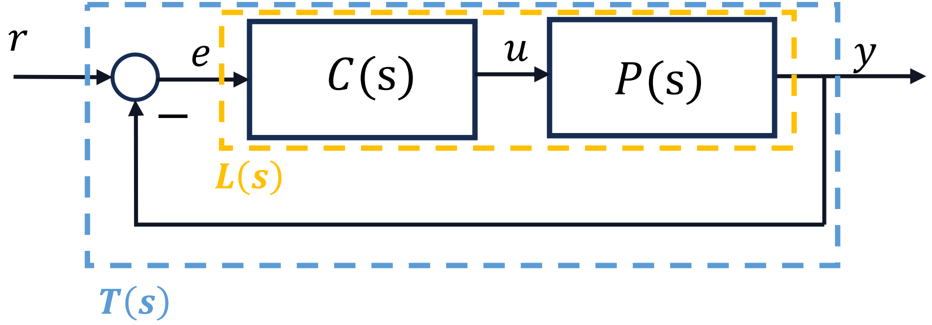
$$y = CPe\,,\ e = r - y \rightarrow y = CP(r-y) \rightarrow y + CPy = CPr \rightarrow y = \frac{CP}{1+CP}r$$

We call this TF Complementary Sensitivity $T(s)$

$$\boldsymbol{T(s) = \frac{C(s)P(s)}{1 + C(s)P(s)} = \frac{L(s)}{1 + L(s)}}$$

r: reference
u: input
y: output
e: error

Key Relations



Summarized are the key relations:

Open-Loop TF: $L(s) = P(s)C(s)$ $(e \rightarrow y)$

Sensitivity: $S(s) = \frac{1}{1+L(s)}$ $(r \rightarrow e)$

Complementary Sensitivity: $T(s) = \frac{L(s)}{1+L(s)}$ $(r \rightarrow y)$

Important: Closed loop and open loop poles are not always equal!

$1 + L(s) = 0$ gives us poles of closed loop transfer function

Effects of Proportional Control $C(s) = k$

The essential question is now; how does feedback affect our system?

<u>Example:</u>

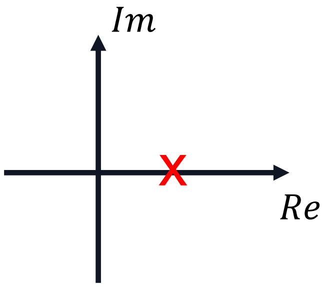
- With the feedback, the open-loop is: $L(s) = \dfrac{k}{s-1}$

$\longrightarrow$ Unstable, no matter the gain!

- If we now close the loop the TF becomes:

$$T(s) = \frac{L(s)}{1+L(s)} = \frac{\frac{k}{s-1}}{1+\frac{k}{s-1}} = \frac{k}{s-1} \cdot \frac{s-1}{(s-1)+k} = \frac{k}{s+k-1}$$

Pole-Zero Plot Open-Loop



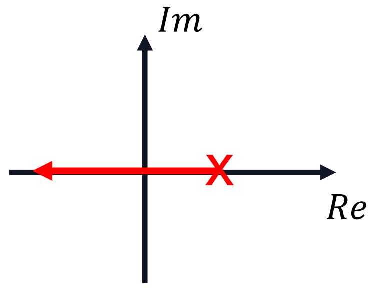
Effects of Proportional Control

- Where are the poles now? ⟶ denominator = 0, $s + k - 1 = 0 \Rightarrow s = 1 - k$

→ It depends on the gain k!

- In this case the closed-loop becomes stable for $k \in (1, \infty)$

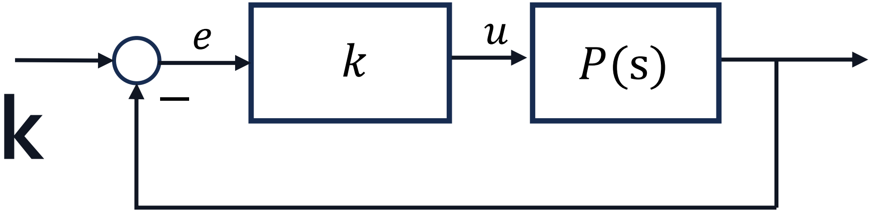
Pole-Zero Plot Closed-Loop



- This is our next important plot…
- A plot that shows the position of poles and zeros over different gains k, is called a **Root-Locus Plot**

2. Root Locus

Proportional Feedback



- We're back with proportional control: $C(s) = k$
- As mentioned earlier we're interested how the system behaves (especially the poles) if we change k

This determines stability

- Root Locus plots show how poles of the closed loop system vary with k
- Let's consider: $T(s) = \frac{kP(s)}{1+kP(s)}$ and $P(s) = \frac{N(s)}{D(s)}$ $\begin{cases} N(s) := \text{Nominator} \\ D(s) := \text{Denominator} \end{cases}$
- For stability let's check the denominator of the closed loop: $1 + k\frac{N}{D} = 0 \Leftrightarrow D + kN = 0$
- If $\boldsymbol{k} = \mathbf{0} \longrightarrow$ poles are determined by $D = 0$, **same as open-loop!**
- If $\boldsymbol{k} = \boldsymbol{\infty} \longrightarrow$ denom. becomes negligible compared to nomin. $\longrightarrow$ **CL-Poles = OL-Zeros**

Let's summarize these rules on the next slide…

Root Locus Rules

Rules for drawing Root Locus

1. The closed-loop poles need to be ***symmetric w.r.t the real axis*** (real or complex conjugate)
2. The degree of $D(s) + kN(s)$ is the same as $D(s)$, which means ***the number of closed-loop poles is the same as the number of open-loop poles***
3. If $k \rightarrow 0$ then $\lim_{k \to 0} D(s) + kN(s) = D(s)$, which means that ***the closed-loop approach the open-loop poles as k $\rightarrow$ 0***.
4. If $k \rightarrow \infty$ and the degree of $N(s)$ is the same as the degree of $D(s)$, then $\lim_{k \to \infty} \frac{D(s)}{k} + N(s) \approx N(s)$, which mean **the closed-loop poles approach the open-loop zeroes**.
5. If the degree of N(s) is smaller, then ***the "excess" closed-loop poles go "to infinity"***

Remember:

$$G(s) = \boldsymbol{k_{rl}} \frac{(s-z_1)(s-z_2)\ldots(s-z_m)}{(s-p_1)(s-p_2)\ldots(s-p_n)}$$

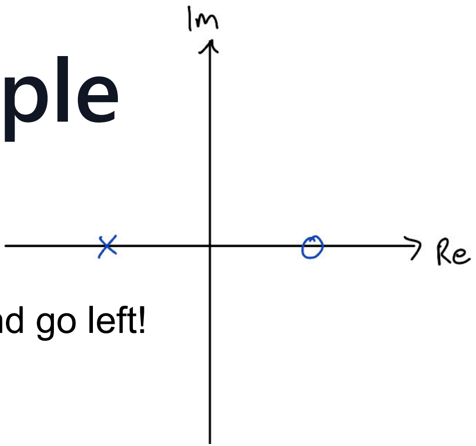
Root-Locus Form

Root Locus Properties

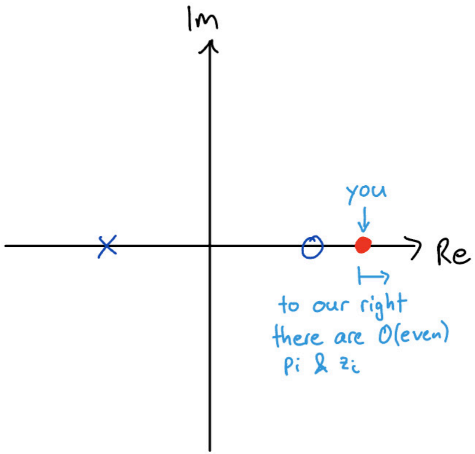
- Root Locus is symmetric w.r.t. the real axis
- If two poles collide, they rotate about 90°
- The locus goes from open-loop poles $(k = 0)$ to open-loop zeros $(|k| \to \infty)$
- For $\boldsymbol{k_{rl}} > 0$: the positive locus lies on the real axis to the **left** of an ***odd*** number of p_i & z_i
- For $\boldsymbol{k_{rl}} < 0$: the negative locus is on the real axis to the **left** of an ***even*** number p_i & z_i (or none)

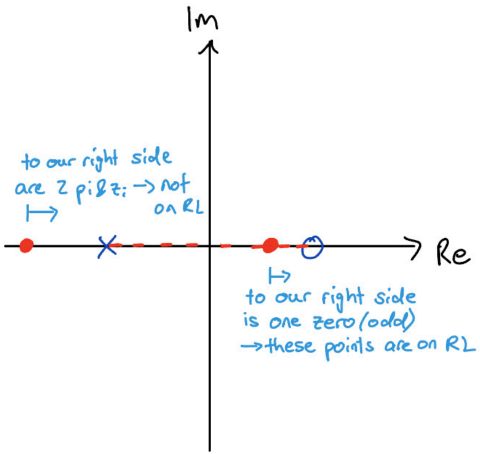
Let's establish these properties visually with an example...

Example

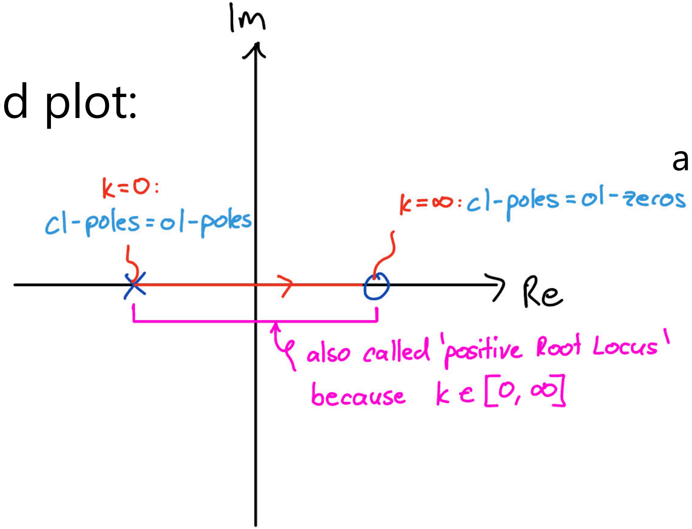


We start right and go left!

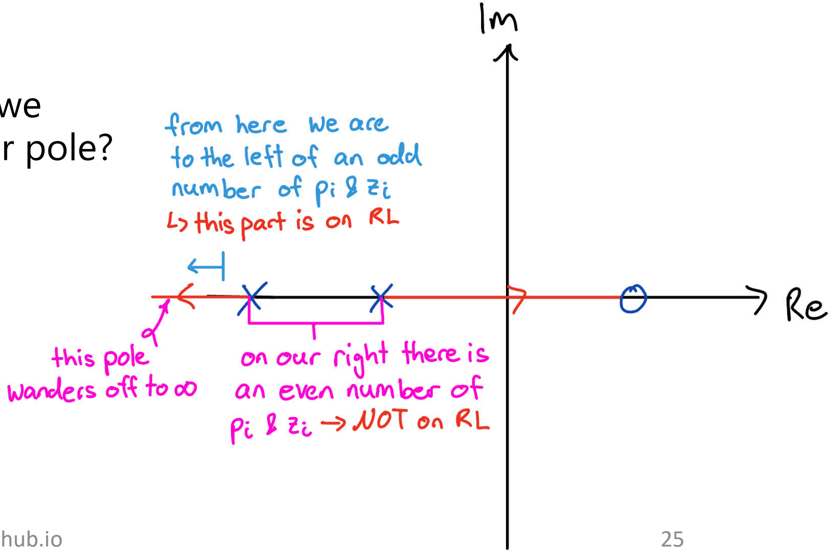




→ finished plot:

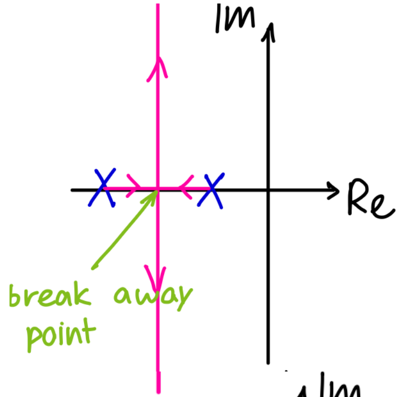


what if we
add another pole?
→



Example

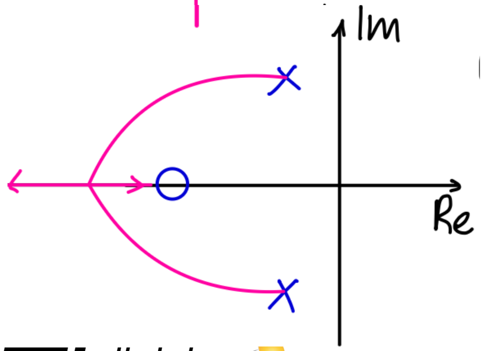
It gets a bit more complicated if two poles collide



from the property before we know that the space between the poles is on the RL

new rule: if two poles collide, they rotate about 90°
⟶ how do they exactly wander off to $\pm\infty$?

⟶ **asymptotes**
⟶ **breakaway points**



another example: one pole wanders off to $-\infty$
and the other one goes to the zero
⟶ what happens in between?

difficult to draw by hand
not relevant to us, mostly
something elliptic looking

Angle and Magnitude Rules

Basic Operation for complex numbers

- $|a \cdot b| = |a| \cdot |b|$
- $|\frac{a}{b}| = \frac{|a|}{|b|}$
- $\angle(a \cdot b) = \angle a + \angle b$
- $\angle(\frac{a}{b}) = \angle a - \angle b$

- Closed loop poles: $D(s) + kN(s) = 0 \rightarrow \frac{N(s)}{D(s)} = -\frac{1}{k}$
- Additional 2 rules, to check if a point is on the Root Locus
- The ***magnitude rule***: Take the magnitude on both sides

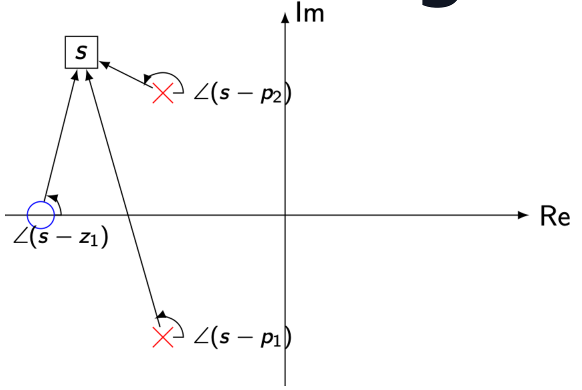
$$\frac{|s - z_1| \cdot |s - z_2| \cdot \ldots \cdot |s - z_m|}{|s - p_1| \cdot |s - p_2| \cdot \ldots \cdot |s - p_n|} = \frac{1}{|k|}$$

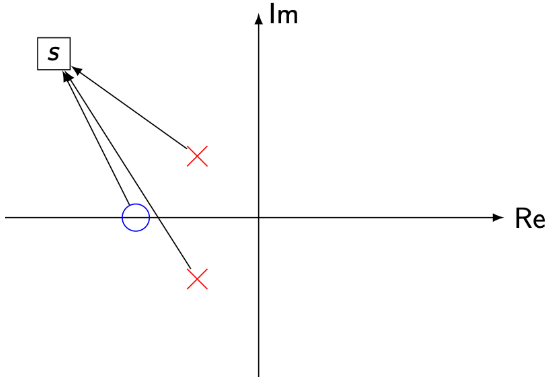
- The ***angle rule***: Take the argument on both side. Note: $q \in \mathbb{N}$

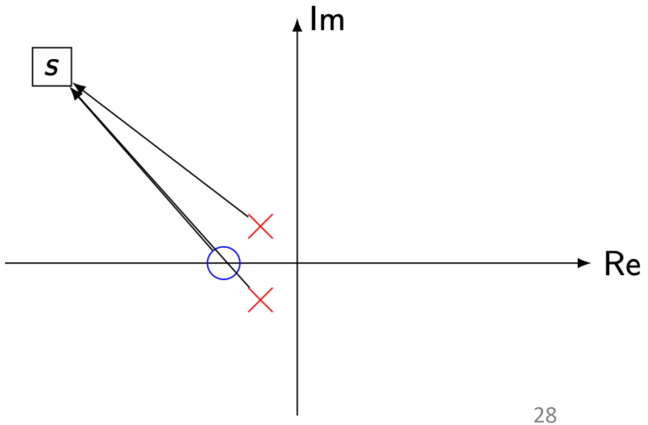
$$\angle(s - z_1) + \cdots + \angle(s - z_m) - \angle(s - p_1) - \cdots - \angle(s - p_n) = \begin{cases} 180^\circ(\pm q \cdot 360^\circ)\ if\ k > 0 \\ 0^\circ(\pm q \cdot 360^\circ)\ if\ k < 0 \end{cases}$$

Zooming Out

- So, what happens when k → ∞ and there are more open-loop poles than zeros?
- We can see, e.g., from the magnitude condition, that the “excess” closed-loop poles will have to go to “to infinity”
- Since this is the complex plane, we need to identify “in which direction” they go towards infinity
- This is where we use the angle rule again







Asymptotes

- If we "zoom out" sufficiently far, the contributions from each of the finite open-loop poles and zeros will all be approximately equal to ∠s, and the angle rule is approximated by $(m-n)\angle s = \angle(-k) \pm q \cdot 360°$
- In other words, as $k \longrightarrow \infty$, the excess poles will go to infinity along asymptotes at angle of

$$\angle s = \frac{180° \pm q \cdot 360°}{n-m} \quad \text{if } k > 0 \qquad\qquad \angle s = \frac{\pm q \cdot 360°}{n-m} \quad \text{if } k < 0$$

- These asymptotes meet in a "center of mass" lying on the real axis at

$$s_{com} = \frac{\sum_{i=1}^{n} p_i - \sum_{j=1}^{m} z_j}{n-m}$$

(poles are "positive" unit masses, and zeros are "negative" unit masses in this analogy)

Now that we have all the tools, this is the final cookbook to follow when sketching the Root Locus

How to Draw Root Locus

1. Calculate poles & zeros of the open-loop and draw them in the complex plane (start & end points of Root Locus)
2. Analyze the real axis (which segments are on RL?)
3. Calculate the asymptotes for the excess poles:

   "center of mass": $$s_{com} = \frac{\sum_{i=1}^{n} p_i - \sum_{j=1}^{m} z_j}{n-m}$$

   Angle: $$\angle s = \frac{180^\circ \pm q \cdot 360^\circ}{n-m} \quad \text{if } k > 0 \qquad \angle s = \frac{\pm q \cdot 360^\circ}{n-m} \quad \text{if } k < 0$$

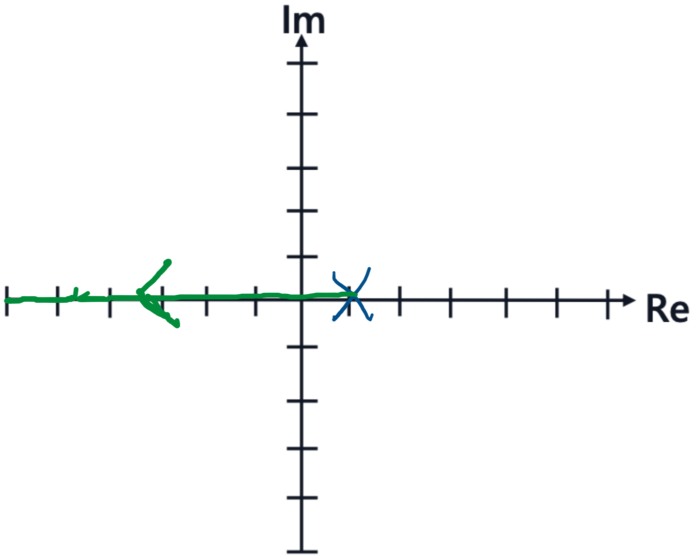
4. Draw the Root Locus

$$\angle s = \frac{\angle(-k) \pm q \cdot 360°}{n - m}$$

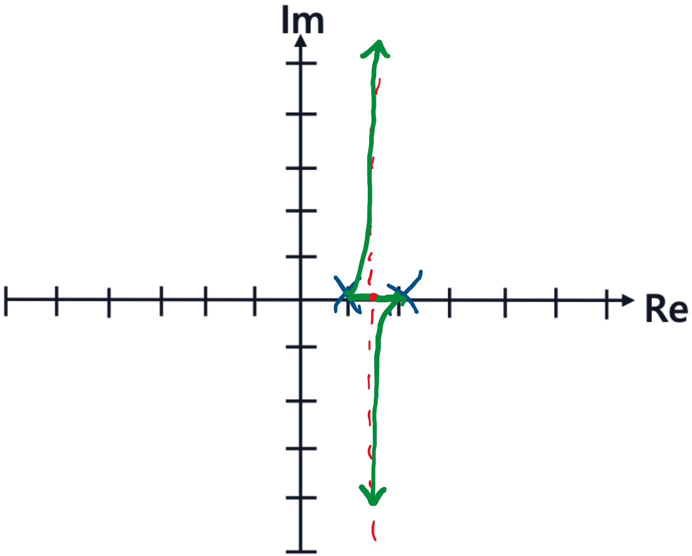
$$s_{\text{com}} = \frac{\sum_{i=1}^{n} p_i - \sum_{j=1}^{m} z_j}{n - m}$$

Examples

$$G_{ol}(s) = \frac{1}{s - 1}$$



$$G_{ol}(s) = \frac{1}{(s-1)(s-2)}$$

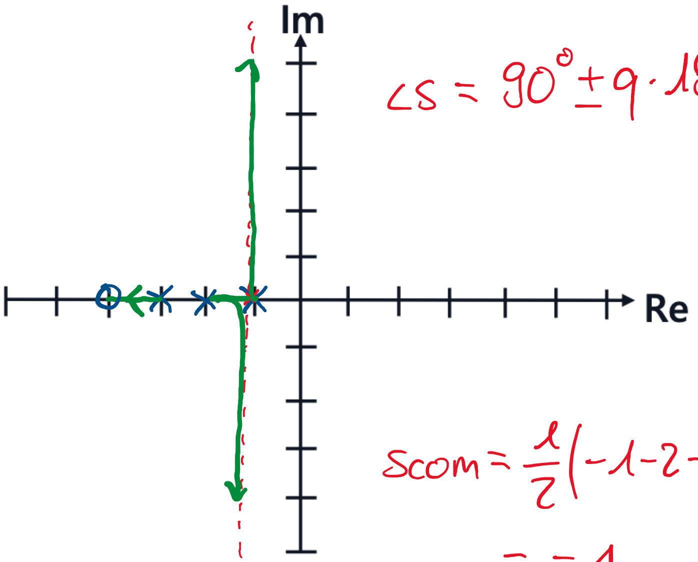


$$\angle s = \frac{\angle(-k) \pm q \cdot 360°}{n - m}$$

$$s_{\text{com}} = \frac{\sum_{i=1}^{n} p_i - \sum_{j=1}^{m} z_j}{n - m}$$

Examples

$$G_{\text{ol}}(s) = \frac{s + 4}{(s + 1)(s + 2)(s + 3)}$$

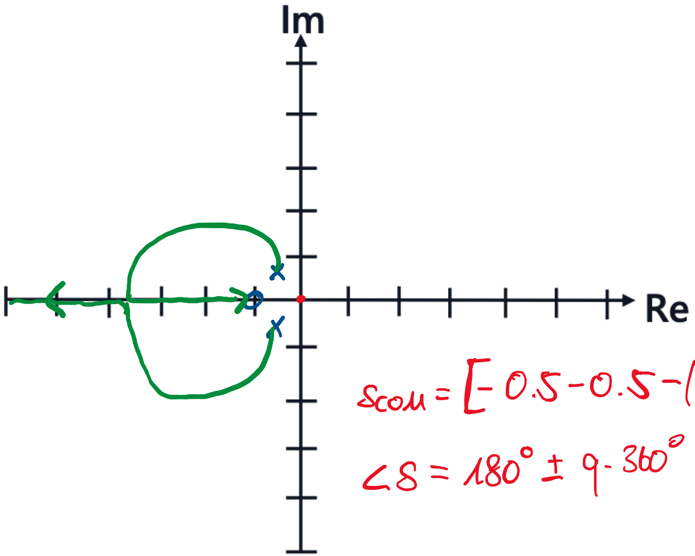


$\angle s = 90° \pm q \cdot 180°$

$s_{com} = \frac{1}{2}(-1-2-3+4)$

$= -1$

$$G_{\text{ol}}(s) = \frac{s + 1}{s^2 + s + 1}$$

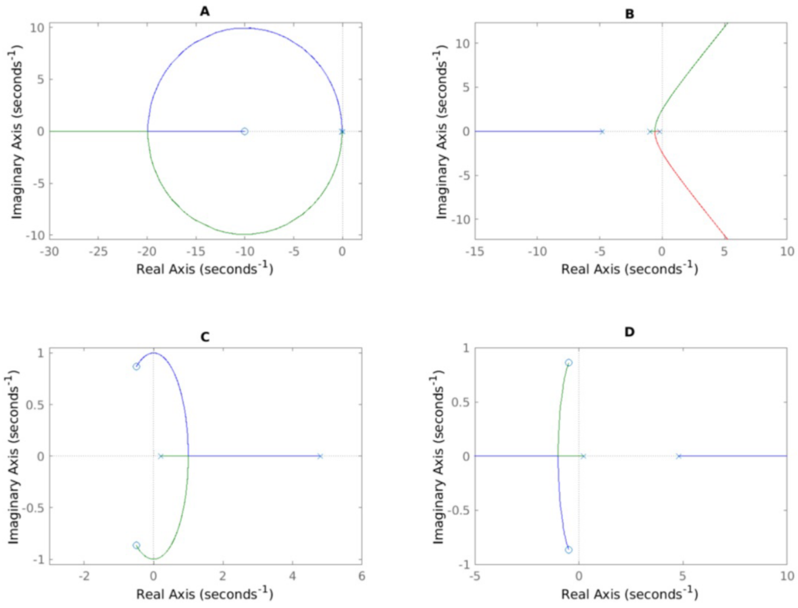


$s_{com} = [-0.5-0.5-(-1)] = 0$

$\angle s = 180° \pm q \cdot 360°$

Exam Question

Problem: Given in Figure 8 are the root locus plots of four different transfer functions for $k > 0$.

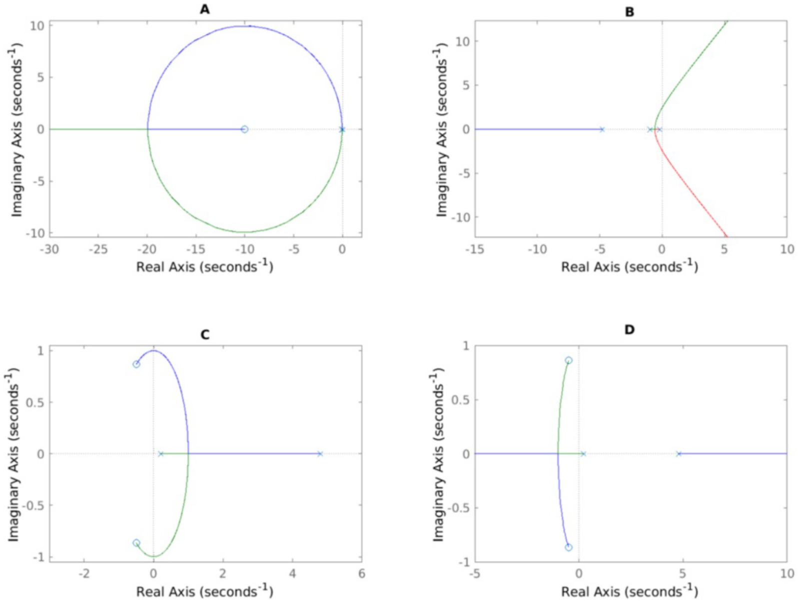


Q22 **(1.5 Points)** Assign each transfer function to the corresponding root locus plot.

Transfer Function	A	B	C	D
$L_1(s) = \dfrac{1}{(s+1)(s^2+5s+1)}$				
$L_2(s) = -\dfrac{s^2+s+1}{s^2-5s+1}$				
$L_3(s) = \dfrac{s+10}{(s+0.01)(s+0.1)}$				
$L_4(s) = \dfrac{s^2+s+1}{s^2-5s+1}$				

Exam Question

Problem: Given in Figure 8 are the root locus plots of four different transfer functions for $k > 0$.



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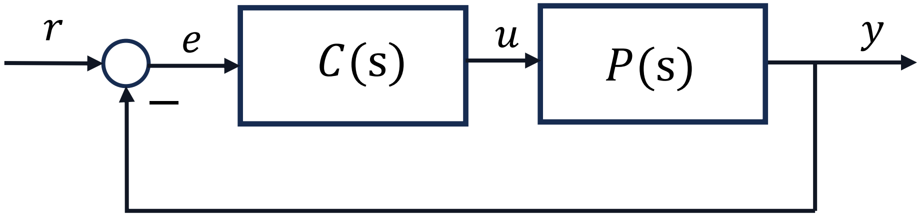
Transfer Function	A	B	C	D
$L_1(s) = \dfrac{1}{(s+1)(s^2+5s+1)}$		X		
$L_2(s) = -\dfrac{s^2+s+1}{s^2-5s+1}$				X
$L_3(s) = \dfrac{s+10}{(s+0.01)(s+0.1)}$	X			
$L_4(s) = \dfrac{s^2+s+1}{s^2-5s+1}$			X	

Different Controllers

- We saw how powerful a simple scalar k as a controller can be
- But most times, our controller will be a dynamic compensator itself!



- A typical $C(s)$ design is combining these three components:
  - Proportional action: k_p
  - Integral action: $k_i \cdot \frac{1}{s}$
  - Derivative action: $k_d \cdot s$

$\longrightarrow$ This is called a PID-controller: $C_{PID}(s) = k_p + k_i \cdot \frac{1}{s} + k_d \cdot s$

I personally would recommend solving at least the problems marked red

Tips for Problem Set 6

Easy	Medium	Hard
3	$1, 2, 5$	$4, 6$

- **Nr. 1:** plug in $u = -ky$ in the state-space form and reorder it to find $\dot{x} = A_{cl}x$
- **Nr. 2:** Especially try to solve a) and d), e). Try understanding the solutions for b) and c) if you don't want to solve it on your own. Focus more on the form of the root locus, instead of the exact breakaway points
- **Nr. 3:** Use magnitude & phase rule
- Nr. 4: Calculations get messy, try to understand solutions
- **Nr. 5:** for a) remember to be Lyapunov Stable: $Re(\lambda_i) \leq 0\ \forall i$

for b) you can solve the whole exercise only looking at open loop p_i & z_i

- Nr.6: look at the slides for the derivation of $T(s)$ and apply it, can give you a good understanding for deriving various transfer functions

Questions?

Feedback?

Too fast? Too slow? Less theory, more exercises?

I would appreciate your feedback. Please let me know.

https://docs.google.com/forms/d/e/1FAIpQLSdHl0kjWo63aNzDkAV0cnmQadCAj5L0D7v7aSh0BK7BBdEgpA/viewform?usp=header